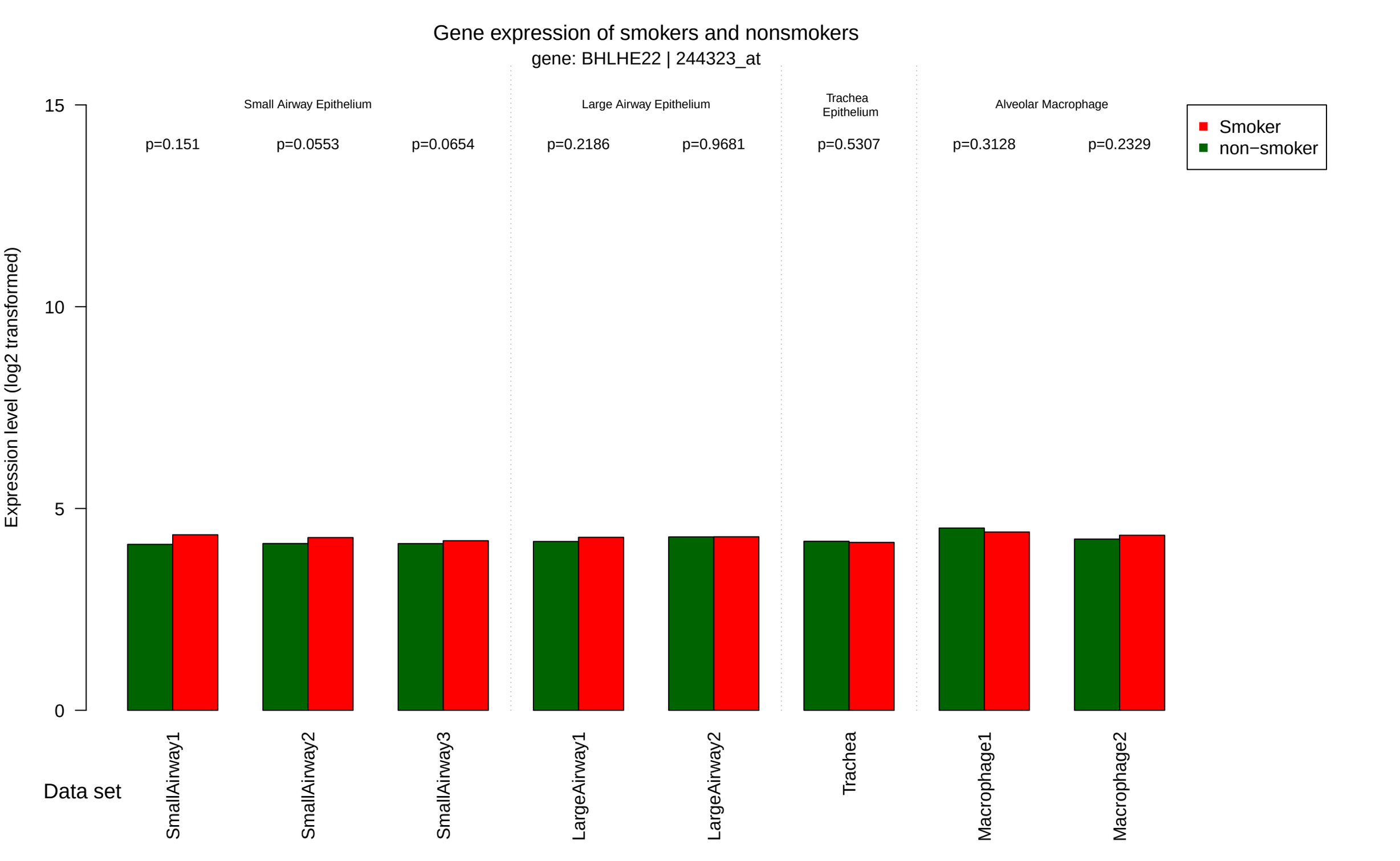
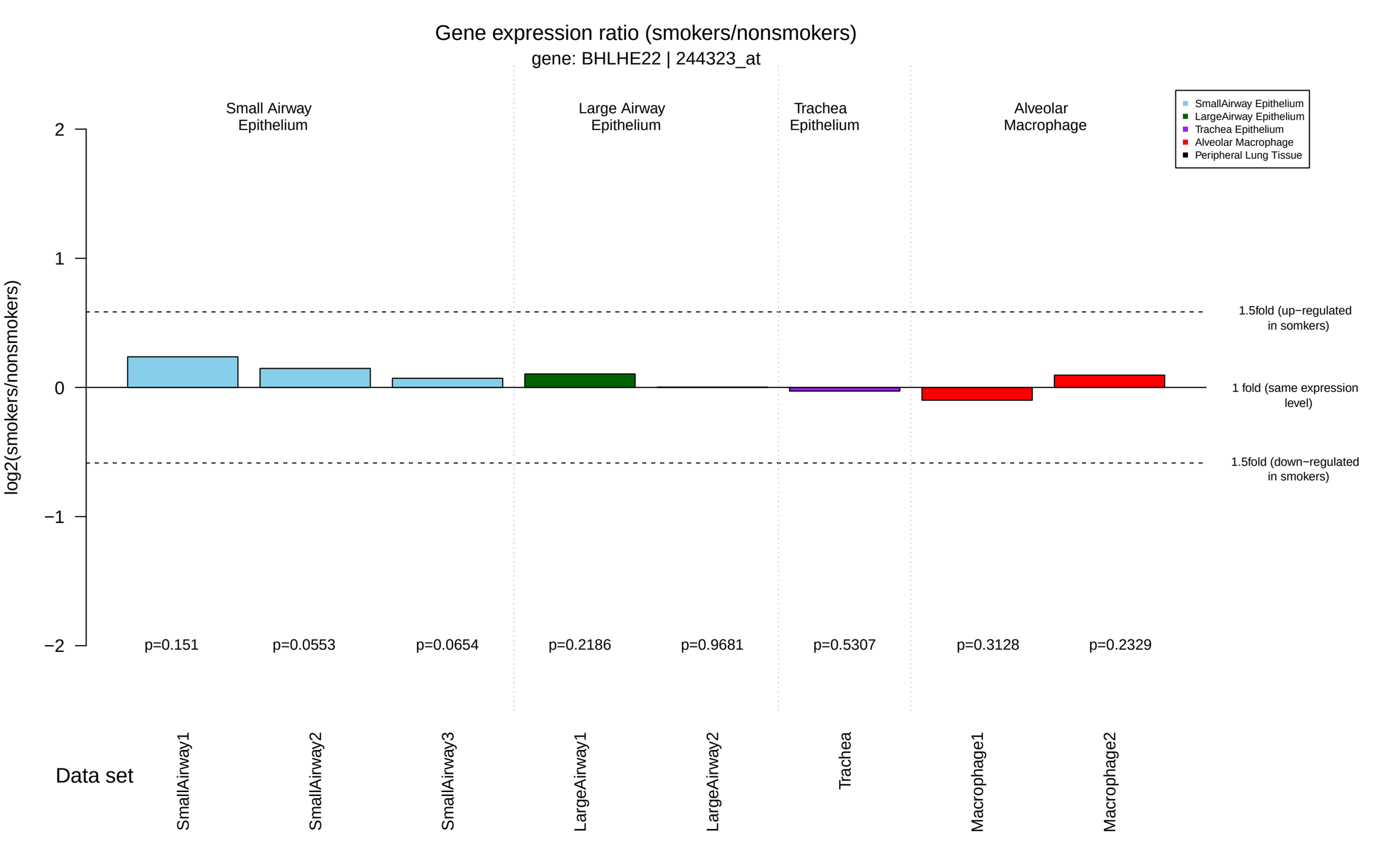
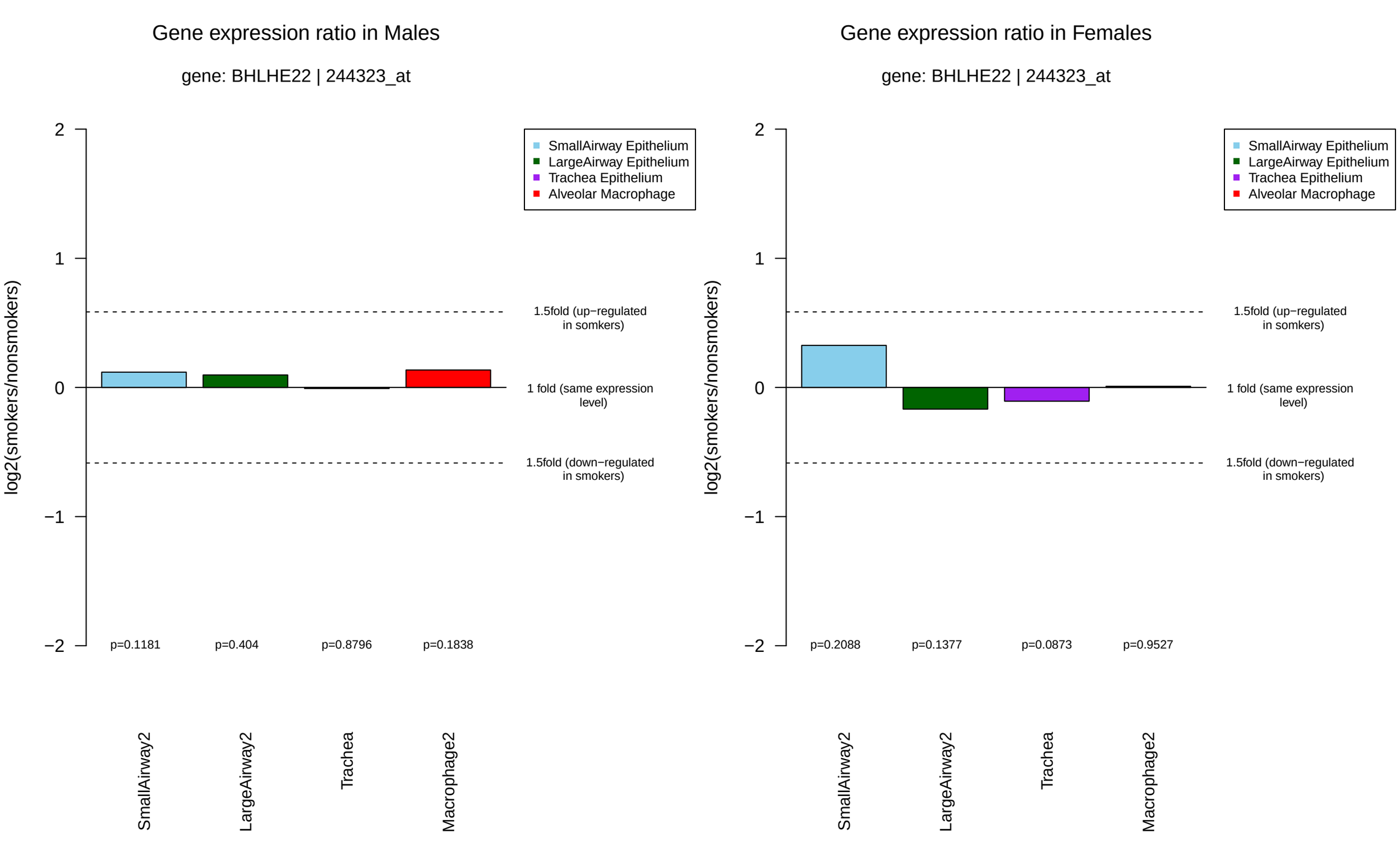
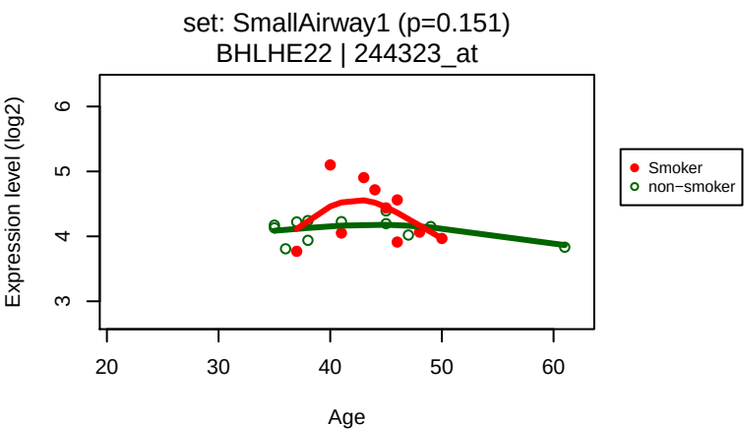
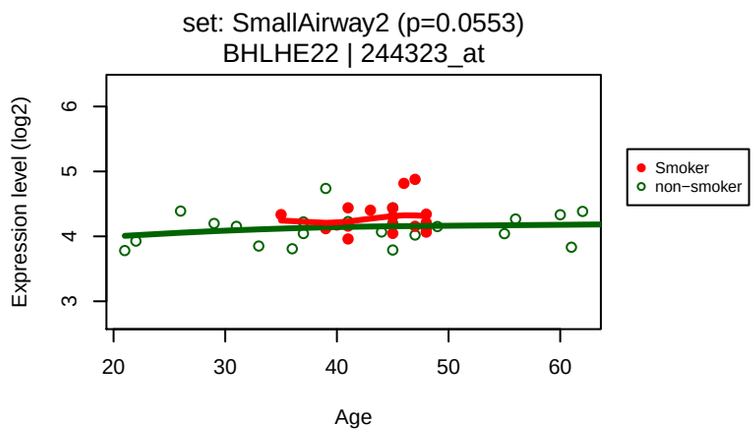
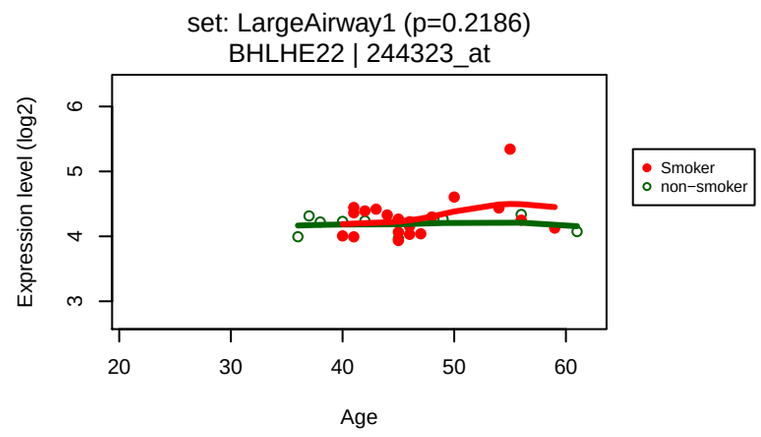

Gene expression of smokers and nonsmokers
gene: BHLHE22 | 244323_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.151
p=0.0553
p=0.0654
p=0.2186
p=0.9681
p=0.5307
p=0.3128
p=0.2329
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

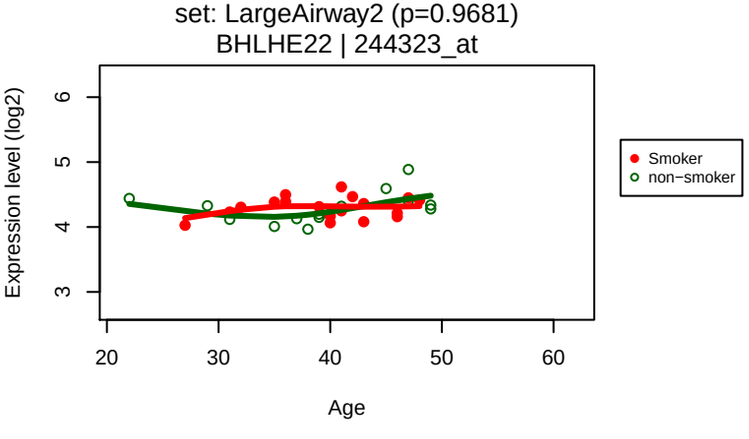

Gene expression ratio (smokers/nonsmokers)
gene: BHLHE22 | 244323_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.151
p=0.0553
p=0.0654
p=0.2186
p=0.9681
p=0.5307
p=0.3128
p=0.2329
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

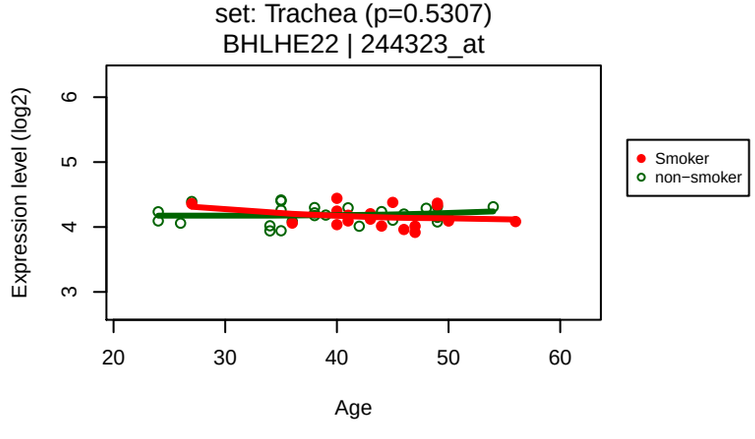

Gene expression ratio in Males
gene: BHLHE22 | 244323_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1181
p=0.404
p=0.8796
p=0.1838
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BHLHE22 | 244323_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2088
p=0.1377
p=0.0873
p=0.9527
SmallAirway2
LargeAirway2
Trachea
Macrophage2

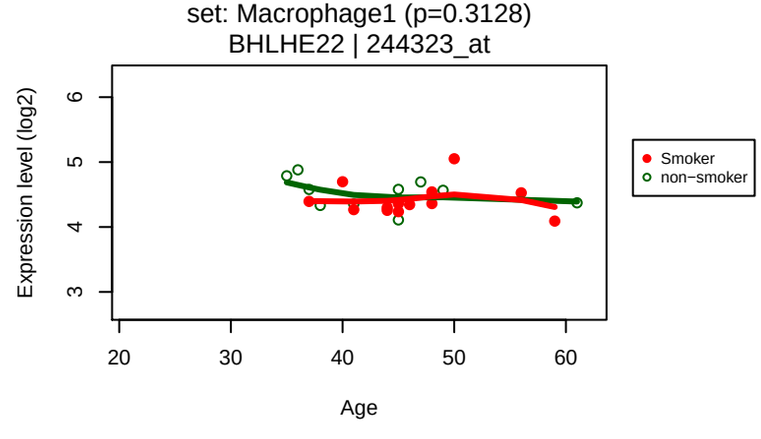

set: SmallAirway1 (p=0.151)
BHLHE22 | 244323_at
Expression level (log2)
Age
Smoker
non−smoker

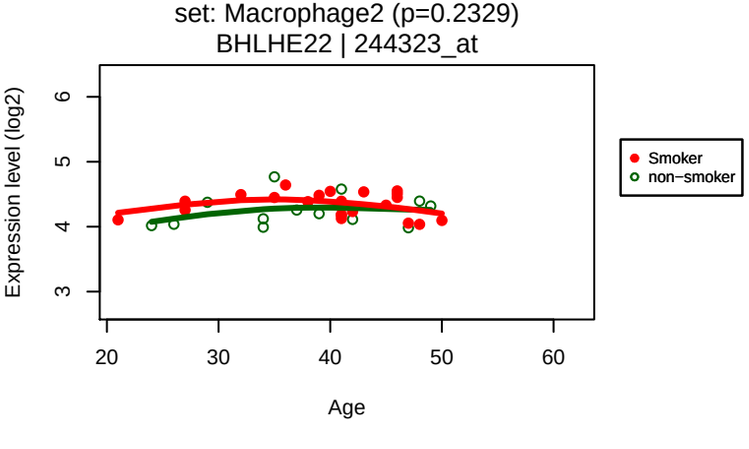

set: SmallAirway2 (p=0.0553)
BHLHE22 | 244323_at
Expression level (log2)
Age
Smoker
non−smoker

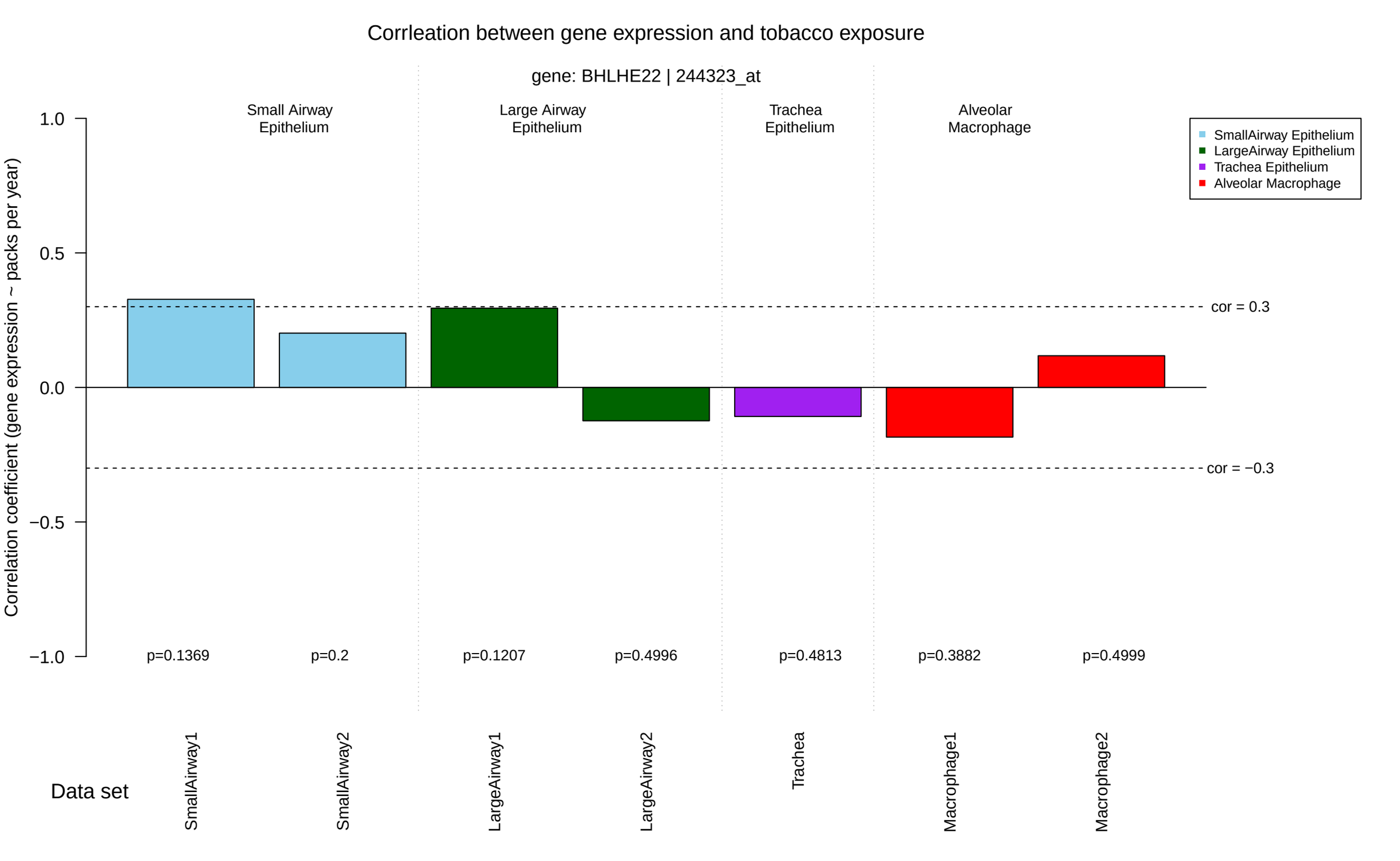

set: LargeAirway1 (p=0.2186)
BHLHE22 | 244323_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9681)
BHLHE22 | 244323_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5307)
BHLHE22 | 244323_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3128)
BHLHE22 | 244323_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2329)
BHLHE22 | 244323_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BHLHE22 | 244323_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1369
p=0.2
p=0.1207
p=0.4996
p=0.4813
p=0.3882
p=0.4999
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2